CUSTOMER RETENTION and CHURN ANALYSIS

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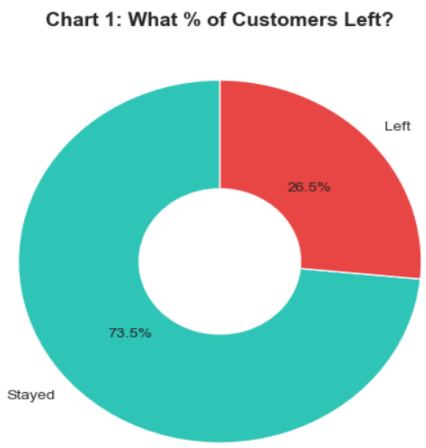
Telco Customer Dataset

Key numbers at glance:

1. Total Customers: 7042
2. Churned: 1869
3. Churn Rate: 26.5%
4. Average Monthly Charge: $64.76
5. Average Tenure: 32.4
6. Average Total Charges: $2283
7. Charts Produced: 11

1.What Percentage of Customers Left



Observation:

1.26.5% of all customers churned, meaning 1 in every 4 customers left

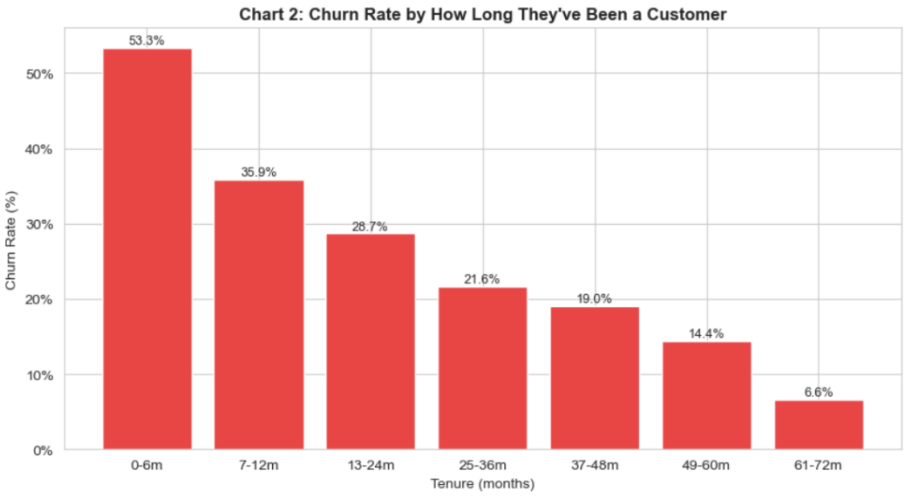
2.In SaaS and subscription businesses, any churn rate above 5 to 7% annually is considered a red flag

3.At 26.5%, the company is losing a quarter of its customer base per cycle

Business Recommendation:

1.Before investing in customer acquisition, fix retention

2. Churn Rate by Tenure (when do customers leave)



Observation:

1.Customers in their first 0–6 months have the highest churn rate, above 45%.

2. Churn drops dramatically after 12 months

3. This is the classic 'early churn curve' where customers who haven't yet found value in the product leave

4. Customers who survive past 2 years are your most loyal

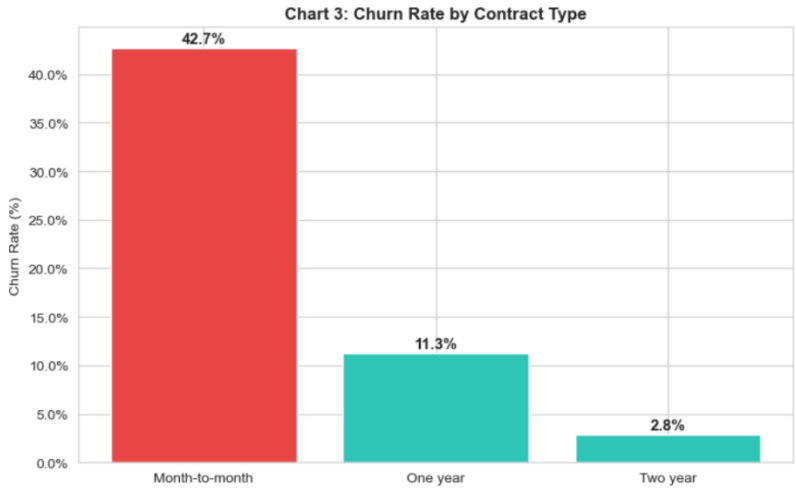
Business Recommendation:

1.The first 90 days are your most critical retention window thus invest in structured onboarding.

2.Send proactive check-in calls or emails at Day 30, Day 60 and Day 90.

3.Introduce milestone rewards like a loyalty discount or free add-on at Month 6.

3.Churn Rate by Contract type



Observation:

1. Month-to-month customers (42.7%) churn the highest of any segment.

2.One-year contract customers (11.3%) have the second highest churn rate

3.Two-year contract customers (2.8%) have the lowest churn rate, nearly 15times lower than month-to-month.

4.Contract type is the single strongest predictor of whether a customer stays or leaves.

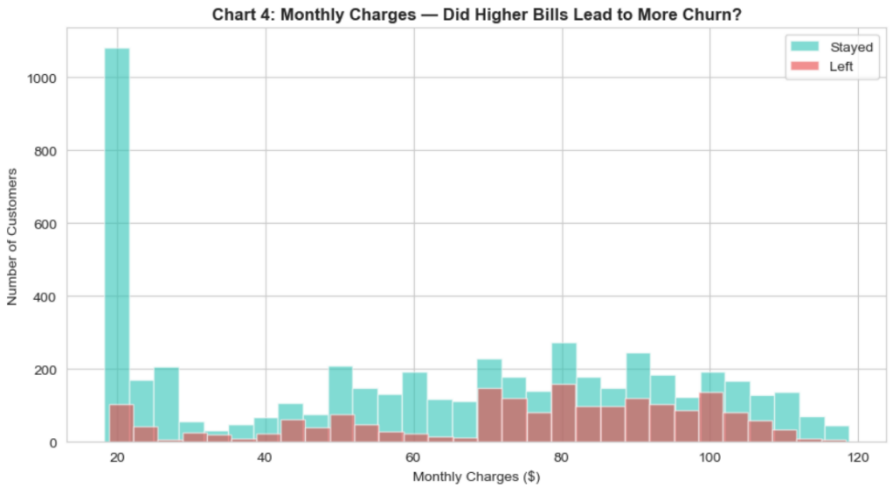
Business Recommendation:

1.Introduce a contract upgrade flow at Month 2, offer 1 free month or a 15% discount to switch to annual.

2.Reframe annual plans as 'loyalty pricing' rather than a commitment.

3.Track contract type as a churn risk flag in your CRM and trigger sales outreach accordingly.

4.Monthly Charges (Do higher bills lead to more churn)



Observation:

1.Churned customers (red) cluster at higher monthly charge levels ($70–$100+).

2.Retained customers (teal) are more evenly distributed, with a larger share at lower price points.

3.High charge customers are not leaving because of the price, they are leaving because they don't feel the value matches what they pay.

4.This is a value perception problem not a pricing problem.

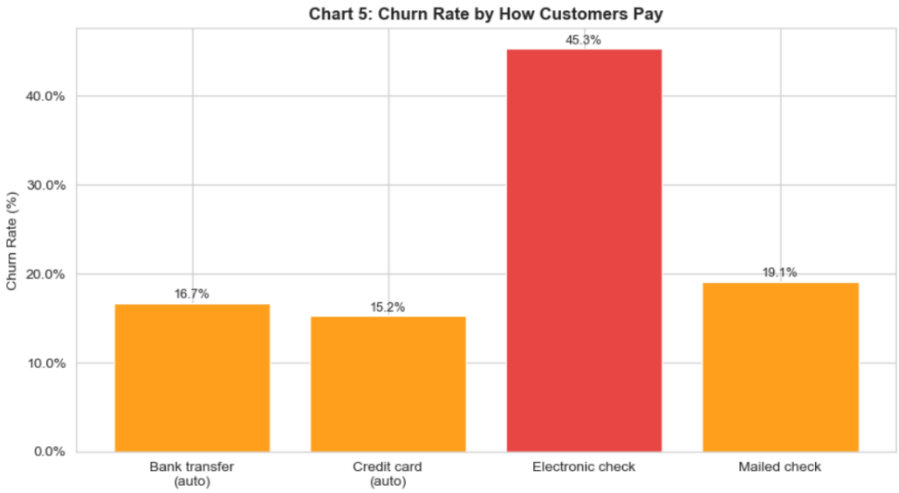
Business Recommendation:

1.High tier customers need proactive success management therefore assign a customer success rep to premium accounts.

2.Introduce a loyalty reward at the 6-month mark for customers on premium plans.

3.Review whether your most expensive plans deliver clearly differentiated value versus cheaper tiers

5.Churn by payment method (How customers pay predicts if they stay)



Observation:

1.Electronic Check customers (45%) churn, the highest of any payment method.

2.Auto pay customers churn nearly 3 times less than electronic check customers.

3. The reason may be that auto pay removes the monthly active decision to keep paying and that mental shift dramatically reduces cancellations

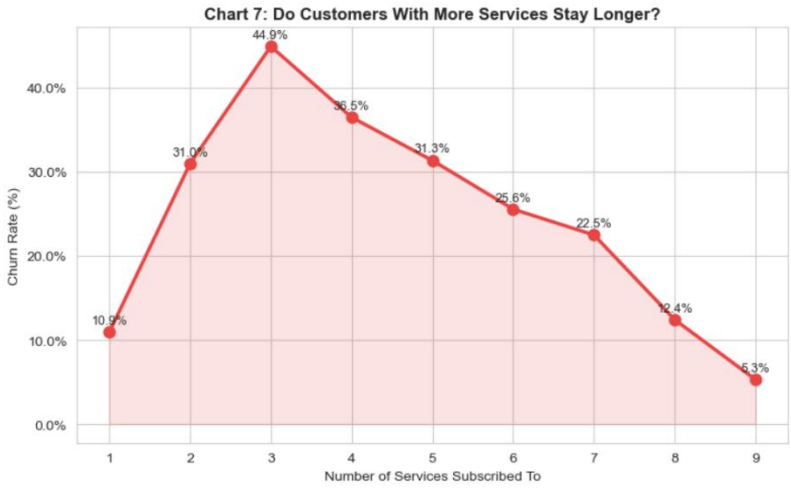
Business Recommendation:

1.Make auto pay enrolment a step in the onboarding flow, not an optional setting buried in account preferences.

2.Offer a small incentive like $5 off your next bill when you switch to auto pay.

3.Flag all electronic check customers as high risk in your CRM for proactive retention outreach.

6.Number of services vs Churn



Observation:

1.Customers with only 3 service have the highest churn rate and there's a visible increase from 1 to 3 number of services subscribed to

2.As the number of services increases, churn rate consistently decreases.

3.Customers embedded deeply in a product ecosystem rarely leave, switching costs become too high.

4.I engineered this 'Service Count' feature during data cleaning by summing across 7 service columns

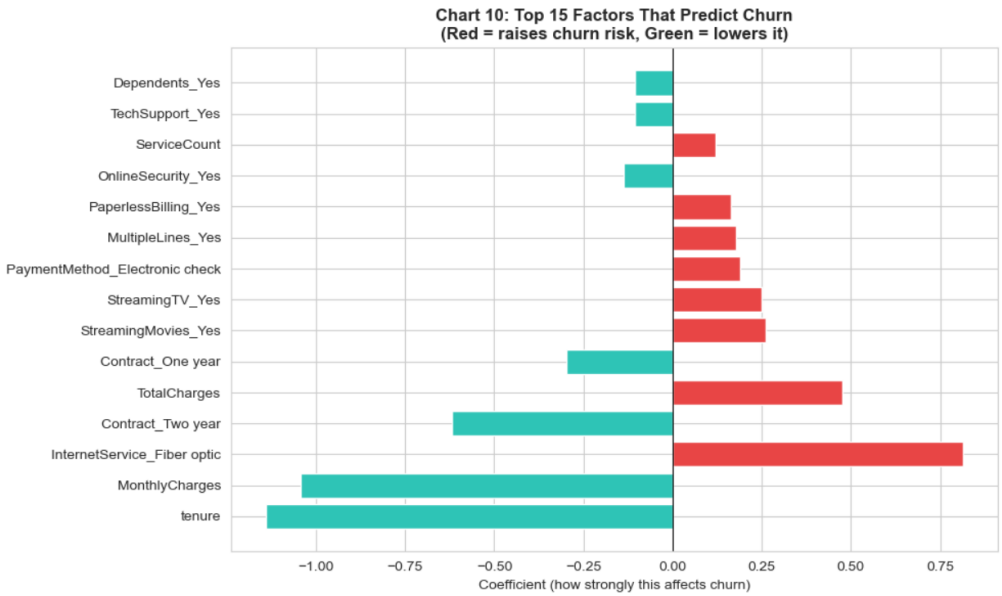
Business Recommendation:

1.Identify customers using only 1–3 services, these are your highest risk cohort.

2.Cross selling is not just a revenue strategy, it is a retention strategy.

3.Track 'service depth' as a leading churn indicator, flag single service customers for outreach

7.Churn Prediction Model (Top 15 risk factors)



Observation:

1.A Logistic Regression model was trained to predict which customers are most likely to churn.

2.The model achieved a ROC-AUC score of 0.83, correctly identifying high risk customers more than 83% of the time.

3.Red bars = factors that INCREASE churn risk:

Month-to-month contract, No Online Security, Fiber Optic, Electronic Check, No Tech Support.

4.Green bars = factors that DECREASE churn risk:

Long tenure, Two-year contract, high service count, having a partner or dependents.